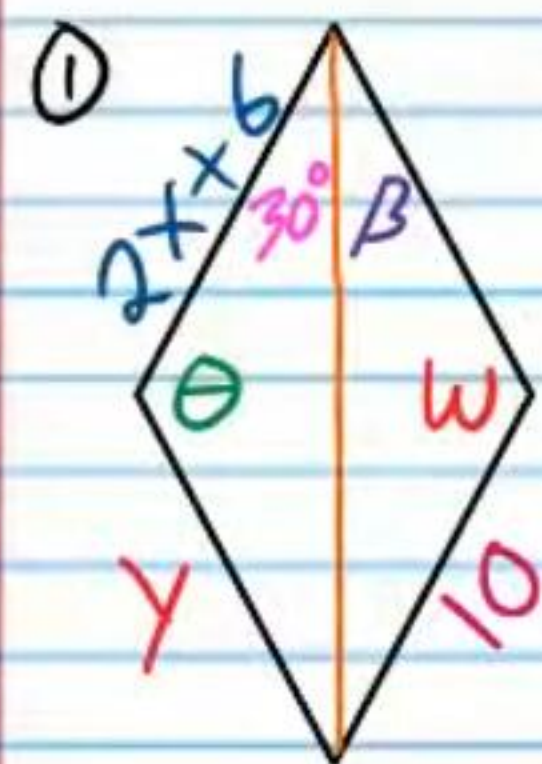
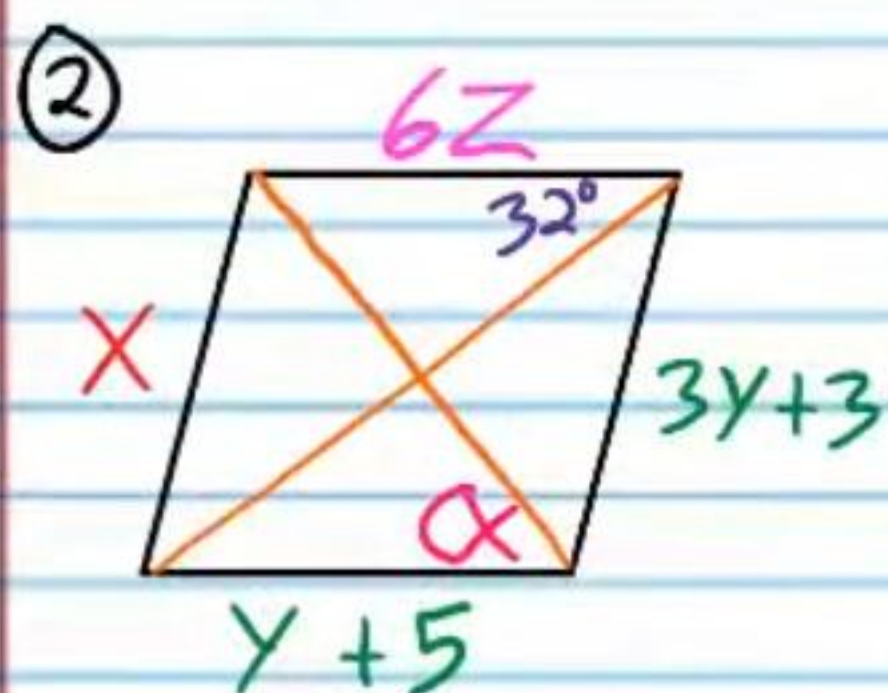


Special Quadrilaterals (Part II): Kites, Rhombi, Trapezoids, Rectangles, and Squares (Homework)

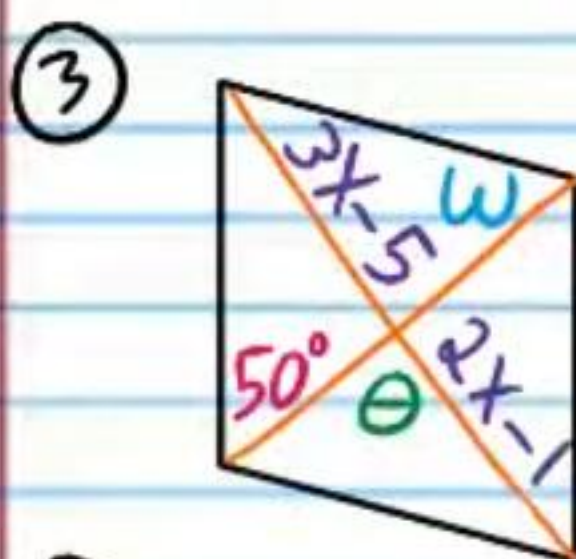
Find the unknowns in each rhombus below.



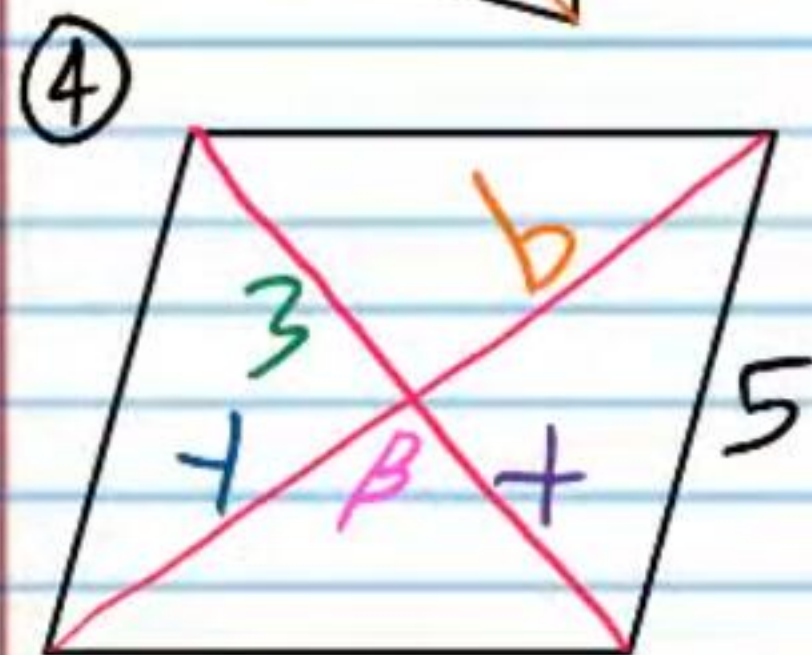
$$\begin{aligned} x &= 2 \\ y &= 10 \\ \beta &= 30^\circ \\ \theta &= 120^\circ \\ w &= 120^\circ \end{aligned}$$



$$\begin{aligned} x &= 6 \\ y &= 1 \\ z &= 1 \\ \alpha &= 58^\circ \end{aligned}$$



$$\begin{aligned} \theta &= 90^\circ \\ w &= 50^\circ \\ x &= 4 \end{aligned}$$

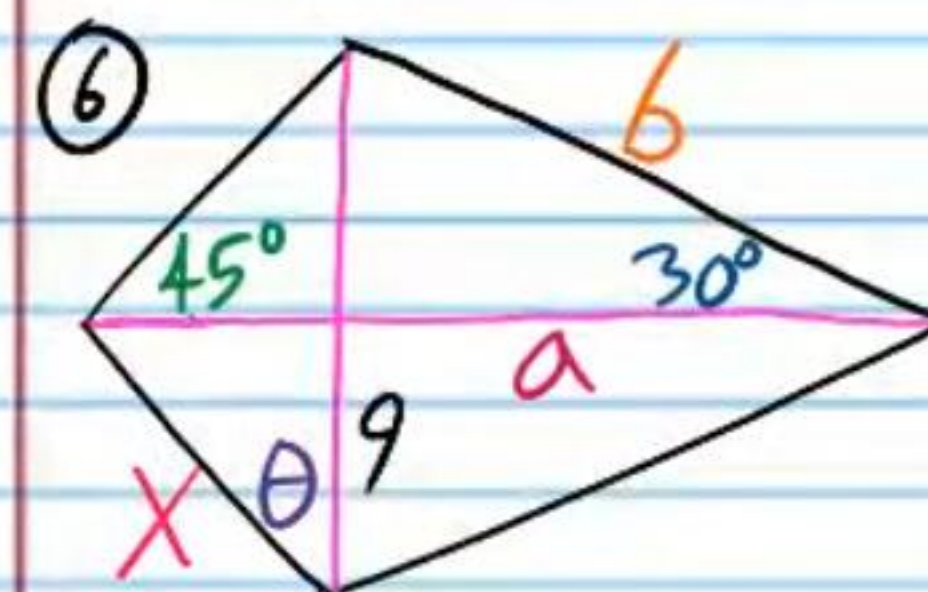


$$\begin{aligned} b &= 4 \\ y &= 4 \\ x &= 3 \\ \beta &= 90^\circ \end{aligned}$$

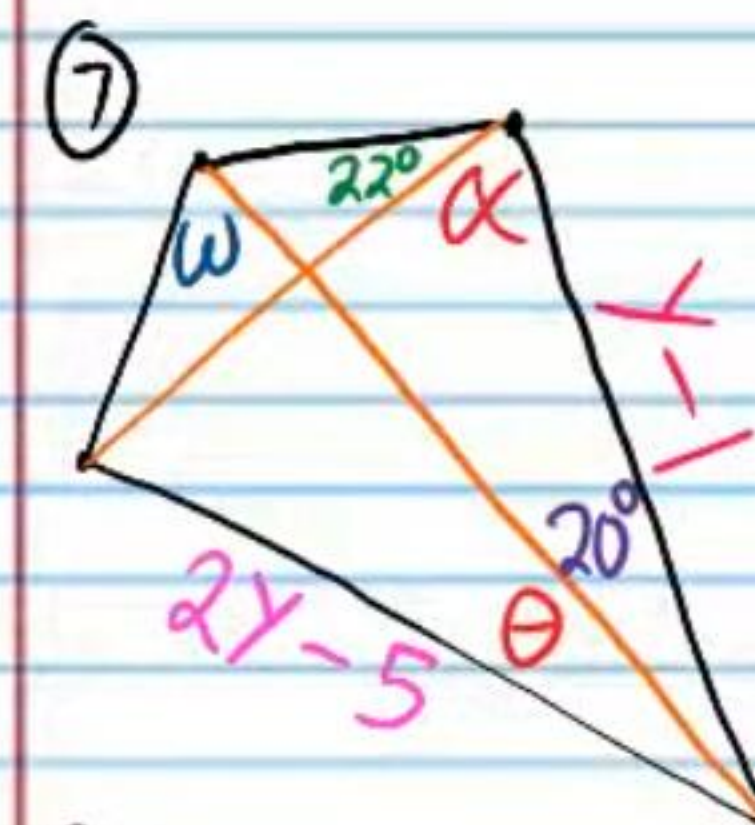
Find the unknowns in each kite below.



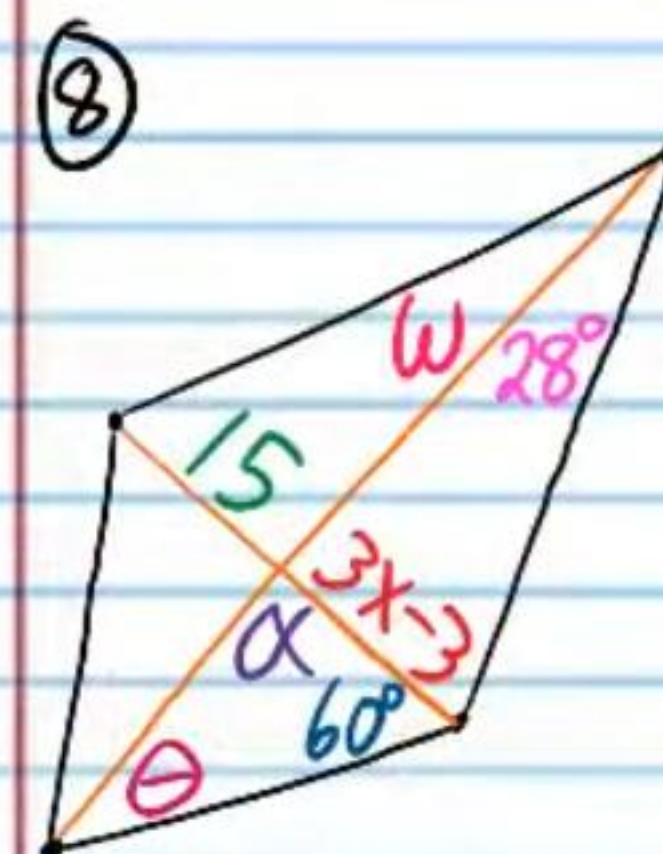
$$\begin{aligned} x &= 3.5 \\ \theta &= 100^\circ \\ \alpha &= 75^\circ \end{aligned}$$



$$\begin{aligned} x &= 9\sqrt{2} \\ b &= 18 \\ a &= 9\sqrt{3} \\ \theta &= 45^\circ \end{aligned}$$



$$\begin{aligned} y &= 4 \\ \alpha &= 70^\circ \\ w &= 68^\circ \\ \theta &= 20^\circ \end{aligned}$$

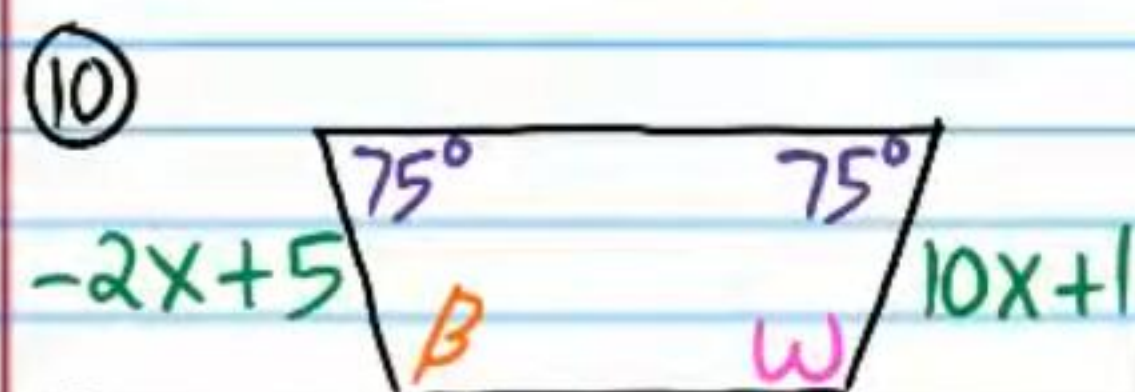


$$\begin{aligned} x &= 6 \\ \alpha &= 90^\circ \\ w &= 28^\circ \\ \theta &= 30^\circ \end{aligned}$$

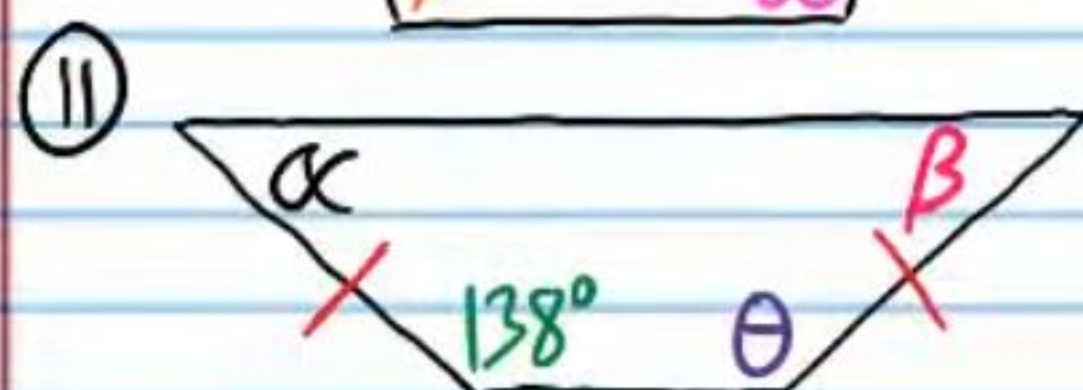
Find the unknowns in each trapezoid below.



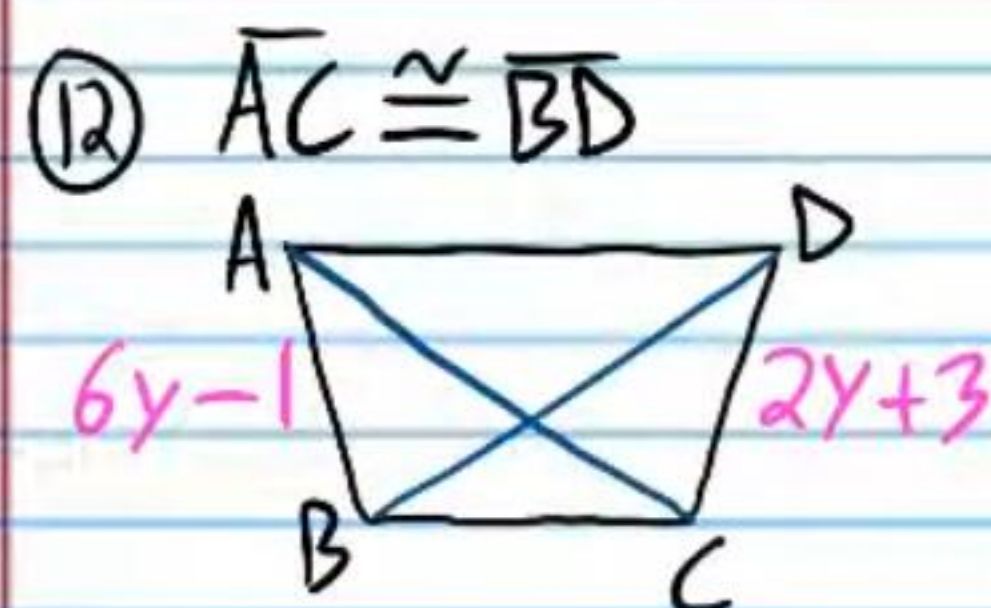
$$\begin{aligned}\theta &= 135^\circ \\ \alpha &= 140^\circ\end{aligned}$$



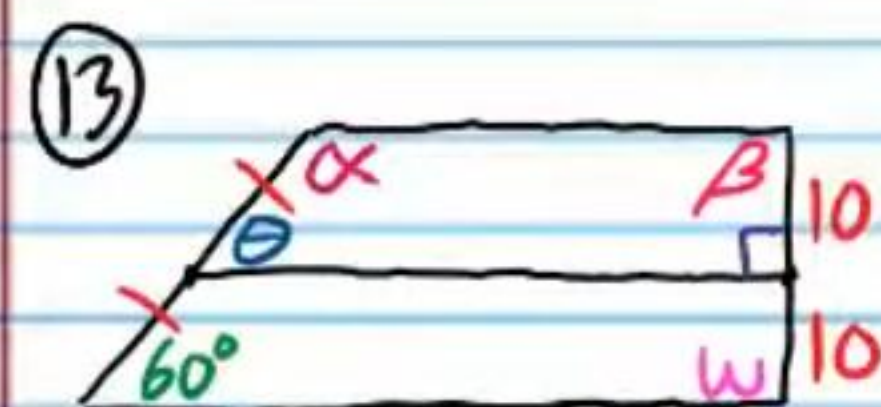
$$\begin{aligned}x &= \frac{1}{3} \\ w &= 105^\circ \\ \beta &= 105^\circ\end{aligned}$$



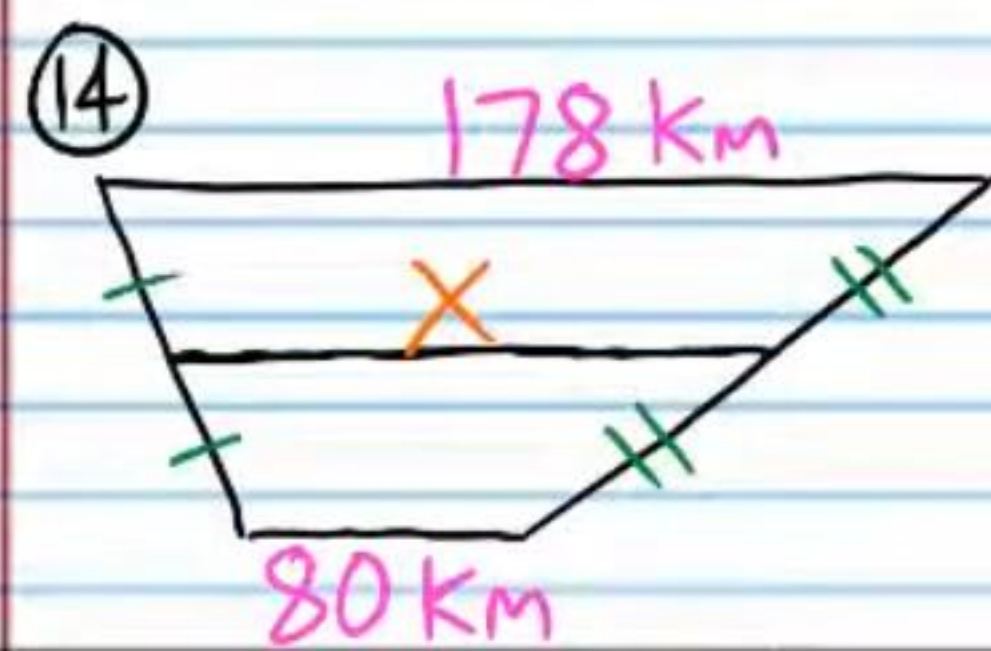
$$\begin{aligned}\theta &= 138^\circ \\ \alpha &= 42^\circ \\ \beta &= 42^\circ\end{aligned}$$



$$y = 1$$



$$\begin{aligned}\theta &= 60^\circ \\ \alpha &= 120^\circ \\ \beta &= 90^\circ \\ w &= 90^\circ\end{aligned}$$

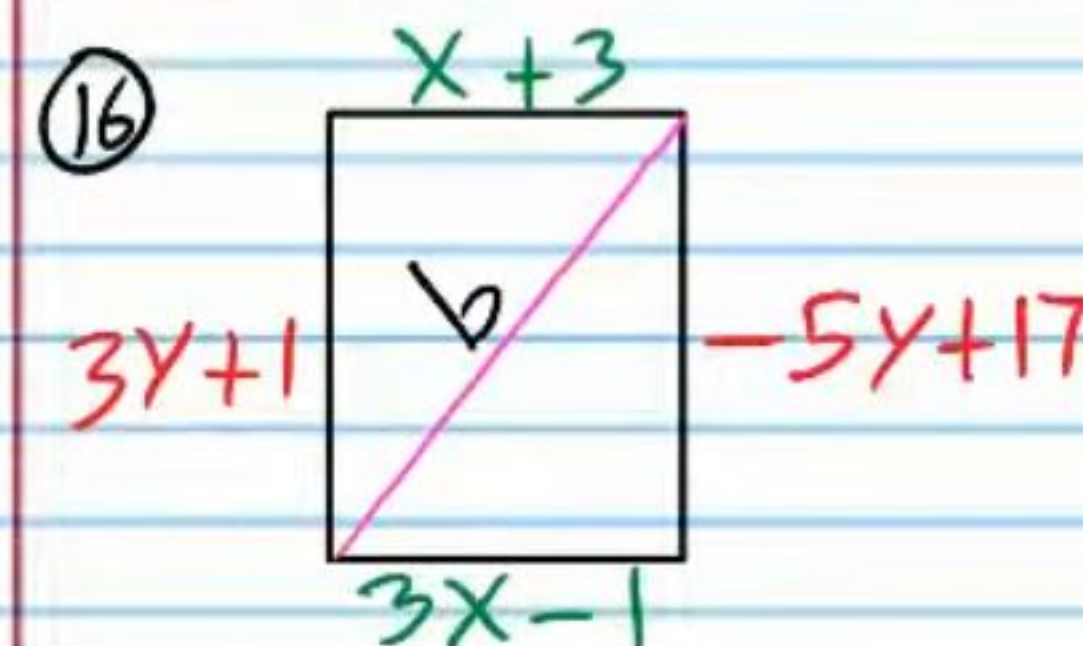


$$x = 129 \text{ km}$$

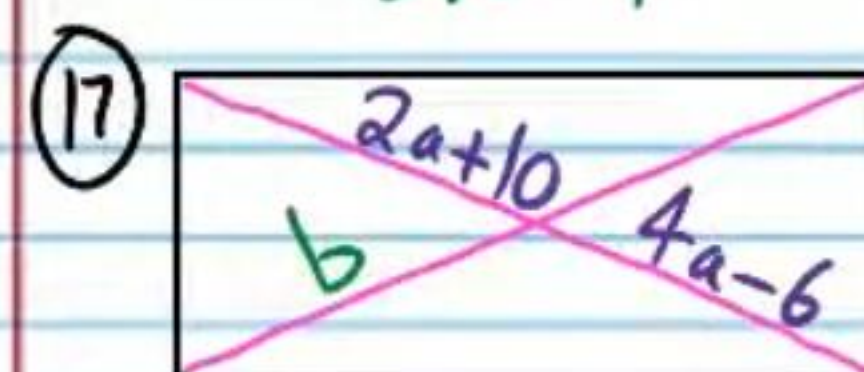


$$x = 6.5$$

Find the unknowns in each rectangle below.

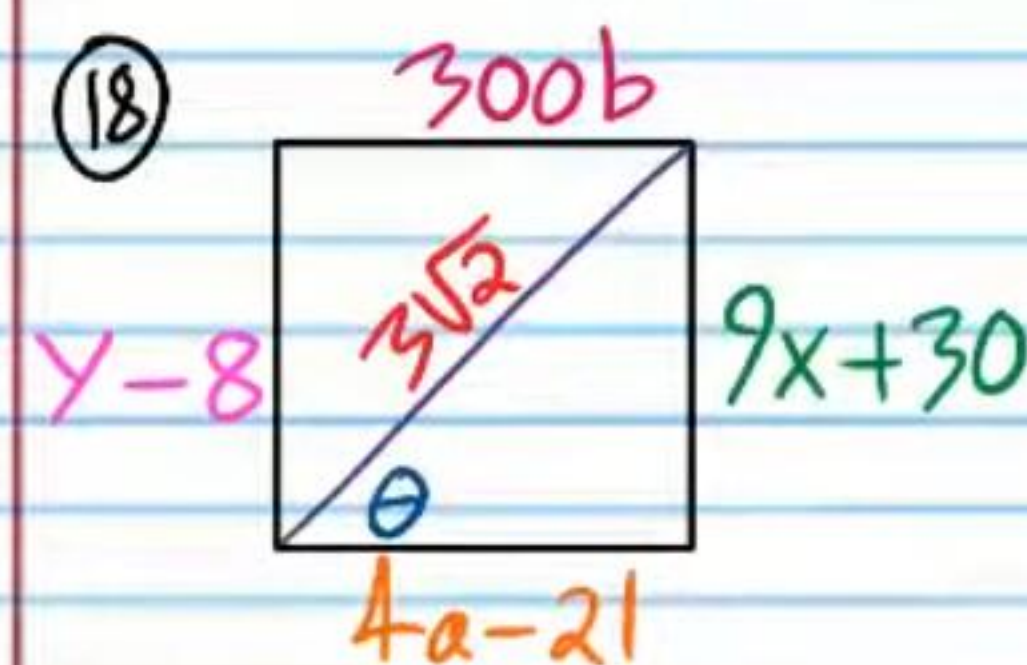


$$\begin{aligned}b &= \sqrt{74} \\ x &= 2 \\ y &= 2\end{aligned}$$



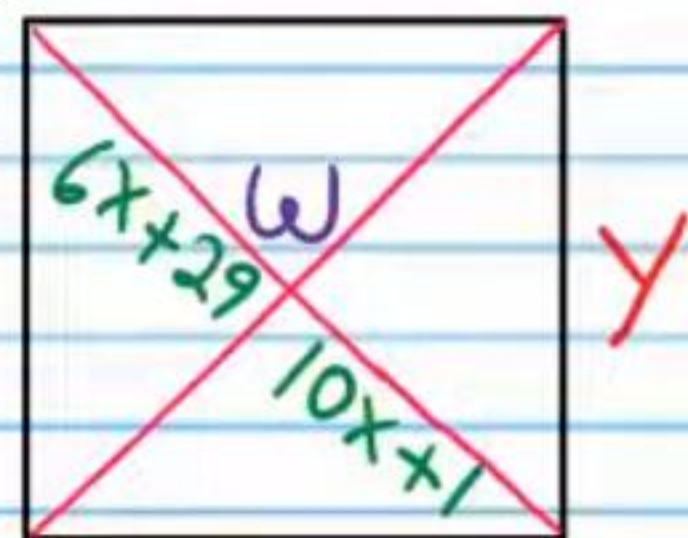
$$\begin{aligned}a &= 8 \\ b &= 26\end{aligned}$$

Find the unknowns in each square below.



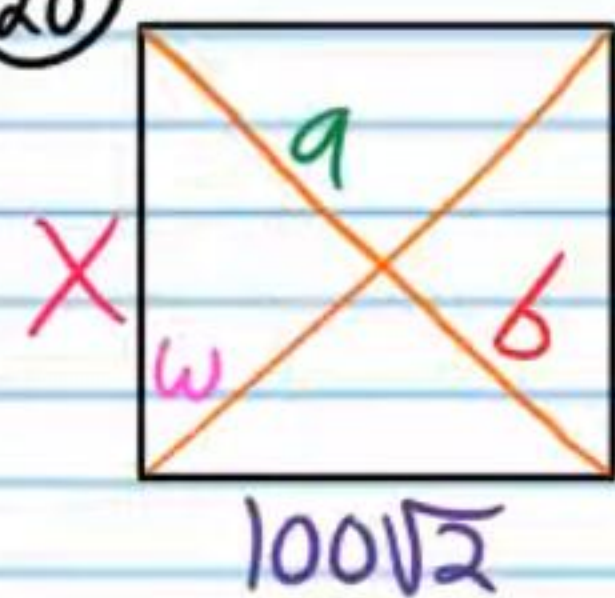
$$\begin{aligned}y &= 11 \\ x &= -3 \\ b &= \frac{1}{100} \\ a &= 6 \\ \theta &= 45^\circ\end{aligned}$$

(19)



$$\begin{aligned} X &= 7 \\ \omega &= 90^\circ \\ Y &= 71\sqrt{2} \end{aligned}$$

(20)



$$\begin{aligned} X &= 100\sqrt{2} \\ a &= 100 \\ b &= 100 \\ \omega &= 45^\circ \end{aligned}$$